
Hechovics

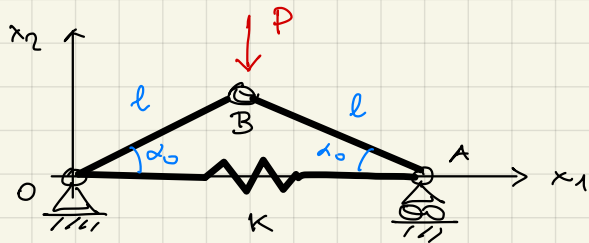
Drionics

lecture 21 (exercises)

7/x11/2022



Exercise 1

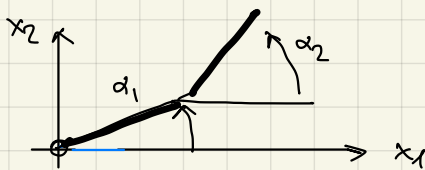


Data:

- OB and AB are rigid rods of length l
- OA is a spring of stiffness K and rest length $L_0 = 2l \cos \alpha_0$
 $V_{el} = \frac{1}{2} K (L - L_0)^2$
 In other words, in the ref. configuration with internal angles α_0 the spring is unextended.
- Pin joint at O, simple support at A, load $-P \underline{e}_2$ at B

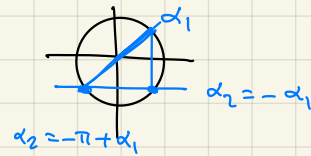
Kinematics

The constraints make this a 1 DoF system.



$$x_2(2l) = l \sin \alpha_1 + l \sin \alpha_2 \stackrel{!}{=} 0 \text{ because of support at A}$$

$$\Rightarrow \sin \alpha_2 = -\sin \alpha_1$$

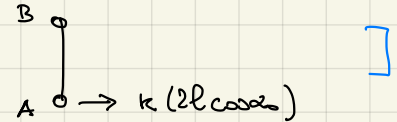


It can be shown that $\alpha_2 = -\pi + \alpha_1$ is never an equilibrium. [SKIP] Actually, $\alpha_2 = -\pi + \alpha_1$ leads to spring of length $L = 0$, "pathological".

$$[\alpha_2 = -\pi + \alpha_1 \Leftrightarrow \alpha_2 - \alpha_1 = \pi]$$



Then $\alpha = \pm \frac{\pi}{2}$ for global equil. and



$M_B(AB)$ cannot vanish

Set $\alpha_1 = \alpha$, $\alpha_2 = -\alpha$. Restrict attention to $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Energy

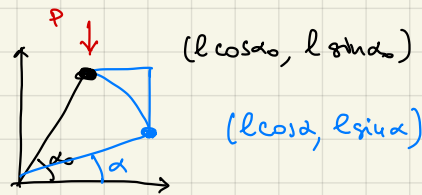
$$V_{el} = \frac{1}{2} K (2l \cos \alpha - 2l \cos \alpha_0)^2$$

$$W_{ext} = -Pl (\sin \alpha - \sin \alpha_0)$$

$$V(\alpha) = V_{el} - W_{ext}$$

$$= 2kl^2 (\cos \alpha - \cos \alpha_0)^2 + Pl (\sin \alpha - \sin \alpha_0)$$

can't



Check: $P \downarrow$ should perform positive work if displacement is downwards, i.e., if $\sin \alpha < \sin \alpha_0$
 $P l (\sin \alpha_0 - \sin \alpha) > 0$ ✓

If in doubt: vertical comp of displacement from $\underline{x}^0$ to $\underline{x}$ is $l \sin \alpha - l \sin \alpha_0$

$$W_{ext} = (-P \underline{e}_2) \cdot (\underline{x}_A(B) - \underline{x}^0(B)) = -Pl (\sin \alpha - \sin \alpha_0)$$

$$V(\alpha) = 2kl^2(\cos\alpha - \cos\alpha_0)^2 + Pl \sin\alpha$$

Equilibria

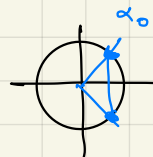
$$\frac{dV}{d\alpha} = 4kl^2(\cos\alpha - \cos\alpha_0)(-\sin\alpha) + Pl \cos\alpha$$

$$\frac{dV}{d\alpha} = 0 = -4kl^2(\cos\alpha - \cos\alpha_0) \sin\alpha + \underbrace{Pl \cos\alpha}_{\neq 0} \Rightarrow P_{eq}(\alpha) = 4kl(\cos\alpha - \cos\alpha_0) \tan\alpha \quad (*)$$

for $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Equilibria for $P=0$: $\alpha=0$, $\alpha=\pm\alpha_0$

Moreover $P_{eq}(\alpha) \rightarrow \begin{matrix} -\infty \\ +\infty \end{matrix}$ for $\alpha \rightarrow \begin{matrix} +\pi/2 \\ -\pi/2 \end{matrix}$



$\cos\alpha = \cos\alpha_0$
 $\rightarrow \alpha = \pm\alpha_0$

If α_0 is small, consider $\alpha \in (-\alpha_0, \alpha_0)$ by expanding $P_{eq}(\alpha)$ around $\alpha=0$

$$\cos\alpha \approx \left(1 - \frac{\alpha^2}{2} + \frac{\alpha^4}{4!} + \dots\right)$$

$$\Rightarrow \cos\alpha - \cos\alpha_0 = \frac{\alpha_0^2}{2} - \frac{\alpha^2}{2} + O(\alpha^4)$$

$$\sin\alpha \approx \left(\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} + \dots\right)$$

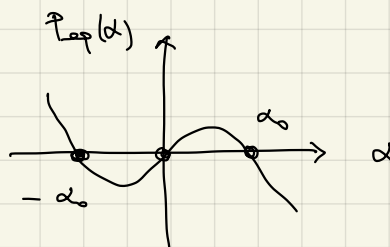
$$\Rightarrow \sin\alpha - \sin\alpha_0 = \alpha - \alpha_0 + O(\alpha^3)$$

$$\tan\alpha \approx \alpha + O(\alpha^3)$$

$$\left[\tan\alpha = 0 + \frac{1}{\cos\alpha} \left| \alpha + (-2)\cos^3\alpha(-\sin\alpha) \right| \frac{\alpha^2}{2} + O(\alpha^3) \right]$$

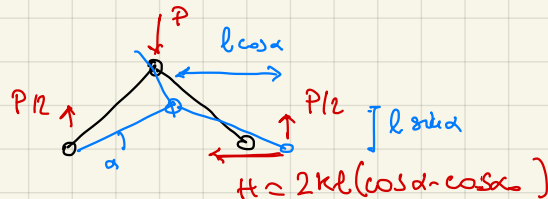
$\underbrace{\frac{1}{\cos\alpha}}_{\alpha=0} = 1 \quad \underbrace{(-2)\cos^3\alpha(-\sin\alpha)}_{\alpha=0} = 0$

$$P_{eq}(\alpha) = 4kl \left(\frac{\alpha_0^2}{2} - \frac{\alpha^2}{2} + \dots \right) (\alpha + \dots) \approx 2kl \underbrace{(\alpha_0^2 - \alpha^2)}_{\approx 2(\cos\alpha - \cos\alpha_0)} \underbrace{\alpha}_{\approx \tan\alpha} \quad \text{approx. by cubic polynomial}$$



Remark

(*) is the relation $P = 2H \tan\alpha \Leftrightarrow \frac{H}{P} = \frac{1}{2} \frac{1}{\tan\alpha}$



$$m_B(BA) = \frac{P}{2} l \cos\alpha - H l \sin\alpha = 0$$

$$\Leftrightarrow \frac{P}{2} \cos\alpha = H \sin\alpha$$

The approximate equil. eq. can be obtained by expanding V_{el} and W_{ext}

$$V_{el} = 2kl^2(\cos\alpha - \cos\alpha_0)^2 \approx 2kl^2\left(\frac{\alpha_0^2}{2} - \frac{\alpha^2}{2}\right)^2 = \frac{1}{2} kl^2(\alpha^2 - \alpha_0^2)^2$$

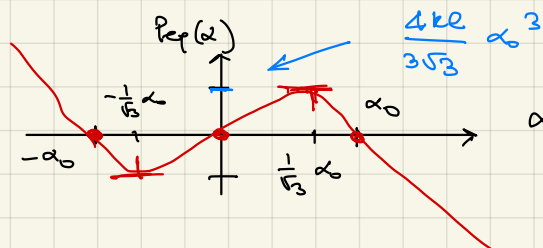
$$W_{ext} = -Pl(\sin\alpha - \sin\alpha_0) \approx -Pl(\alpha - \alpha_0)$$

$$V = V_{el} - W_{ext} \approx \frac{1}{2} kl^2(\alpha^2 - \alpha_0^2)^2 + Pl(\alpha - \alpha_0)$$

$$\frac{dV}{d\alpha} = kl^2(\alpha^2 - \alpha_0^2)2\alpha + Pl; \quad \frac{dV}{d\alpha} = 0 \rightarrow P_{eq}(\alpha) = -2kl(\alpha^2 - \alpha_0^2)\alpha$$

$$P(\alpha) = 0 \text{ for } \alpha = 0, \alpha = \pm\alpha_0$$

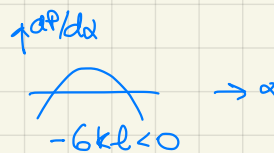
$$P(\alpha) \rightarrow \begin{matrix} -\infty \\ +\infty \end{matrix} \text{ for } \alpha \rightarrow \begin{matrix} +\infty \\ -\infty \end{matrix}$$



$$P(\alpha) = 2kl(\alpha_0^2\alpha - \alpha^3)$$

$$\frac{dP}{d\alpha} = 2kl(\alpha_0^2 - 3\alpha^2) = 0 \text{ for } \alpha = \pm \frac{1}{\sqrt{3}} \alpha_0$$

$$\text{a quadratic polynomial } \begin{cases} > 0 & \text{for } |\alpha| < \frac{1}{\sqrt{3}} \alpha_0 \\ < 0 & \text{for } |\alpha| > \frac{1}{\sqrt{3}} \alpha_0 \end{cases}$$



→ Hint for the fully nonlinear case: $P(\alpha)$ strictly monotone outside of the interval contained in $(-\alpha_0, \alpha_0)$, see below

$$P\left(\pm \frac{1}{\sqrt{3}} \alpha_0\right) = 2kl\left(\alpha_0^2 - \frac{\alpha_0^2}{3}\right)\left(\pm \frac{\alpha_0}{\sqrt{3}}\right) = \pm \frac{4kl}{3\sqrt{3}} \alpha_0^3$$

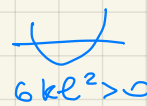
$$\text{from } P_{eq}(\alpha) = 2kl(\alpha_0^2 - \alpha^2)\alpha \rightarrow \frac{3}{3} - \frac{1}{3} = \frac{2}{3}$$

look at sign of $\frac{d^2V}{d\alpha^2}$ for stability

$$\frac{dV}{d\alpha} = 2kl^2(\alpha^3 - \alpha_0^2\alpha) + Pl$$

$$\frac{d^2V}{d\alpha^2} = 2kl^2(3\alpha^2 - \alpha_0^2) \text{ quadratic function}$$

$$= 0 \text{ for } \alpha = \pm \frac{\alpha_0}{\sqrt{3}}$$

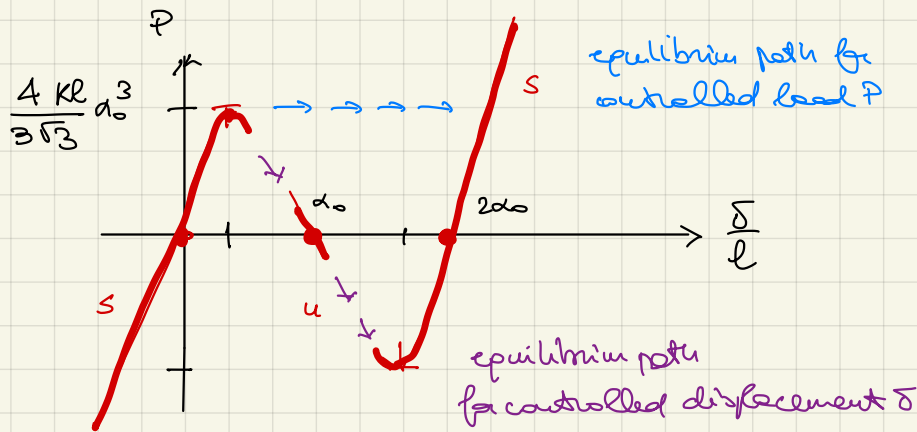
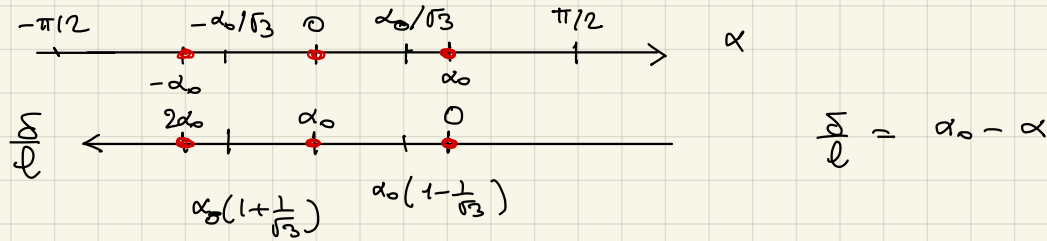


$$\begin{cases} > 0 & \text{for } |\alpha| > \frac{\alpha_0}{\sqrt{3}} & (\text{stable}) \\ < 0 & \text{for } |\alpha| < \frac{\alpha_0}{\sqrt{3}} & (\text{unstable}) \end{cases}$$

Notice correlation between sign of $\frac{d^2V}{d\alpha^2}$ and $\frac{dP}{d\alpha}$

To plot equilibrium response it is useful to change variables $\alpha \rightarrow \frac{\delta}{l}$

where $\delta = -l(\sin\alpha - \sin\alpha_0) \approx -l(\alpha - \alpha_0)$ (downward vertical displacement at B)



Limit-point instability
(snap-through)

Stable equilibria are such that $\left. \frac{d}{d\delta} P_{eq}(\delta) \right|_{\delta_{eq}} > 0$

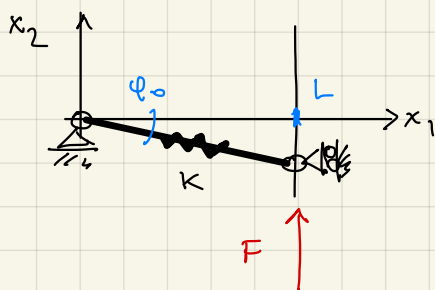
The full analysis, without polynomial expansions:

$\alpha_0(1 + \frac{1}{\sqrt{3}})$ is really $\cos^{-1}((\cos\alpha_0)^{1/3})$

$P(\alpha) = 4kl(\cos\alpha - \cos\alpha_0)$ tends to $\pm\infty$ for $\alpha \rightarrow \pm\pi/2$

Stable equilibria are those for which $\frac{d}{d\delta} P_{eq}(\delta) > 0$

Exercise 2



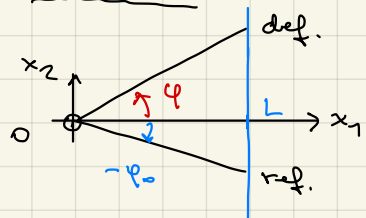
The figure shows the reference configuration for a spring of stiffness K in its rest configuration. Let l_0 be the spring length in that configuration, l its length in a deformed configuration. The energy of the spring is $\frac{1}{2} K (l - l_0)^2$

loads: force $\underline{F} = F \underline{e}_2$, $F \in (-\infty, +\infty)$ applied at right end of the spring

- Compute the equilibrium configurations for $F = 0$
- Using the assumption $\varphi_0 \ll 1$ (small angle) determine whether there are equilibrium configurations for small values of F
- Find approximate expressions for the relation between equilibrium configurations and the corresponding values of F

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Kinematics



$$\varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Def. config. = rest config. for spring: $\varphi = -\varphi_0$, $\varphi_0 \ll 1$

$$\text{Spring length } l(\varphi) = \frac{L}{\cos \varphi}, \quad l_0 = l(-\varphi_0) = \frac{L}{\cos \varphi_0}$$

$$x_2(l, \varphi) = L \tan \varphi$$

Energy

$$V_{el}(\varphi) = \frac{1}{2} K (l(\varphi) - l(-\varphi_0))^2 = \frac{1}{2} K \left(\frac{L}{\cos \varphi} - \frac{L}{\cos \varphi_0} \right)^2$$

$$W_{ext}(\varphi) = F (L \tan \varphi - L \tan(-\varphi_0)) = FL (\tan \varphi + \tan \varphi_0)$$

$$V(\varphi) = \frac{1}{2} KL^2 \left(\frac{1}{\cos^2 \varphi} - \frac{2}{\cos \varphi \cos \varphi_0} + \frac{1}{\cos^2 \varphi_0} \right) - FL (\tan \varphi + \tan \varphi_0)$$

constant

$$\Leftrightarrow \frac{1}{2} KL^2 \left[\frac{1}{\cos^2 \varphi} - \frac{2}{\cos \varphi \cos \varphi_0} \right] - FL \tan \varphi$$

Equilibrium

$$\begin{aligned} \frac{dV}{d\varphi} &= \frac{1}{2} KL^2 \left[(-2) \frac{1}{\cos^3 \varphi} (-\sin \varphi) - \frac{2}{\cos \varphi_0} \left(-\frac{(-\sin \varphi)}{\cos^2 \varphi} \right) \right] - FL \frac{1}{\cos^2 \varphi} \\ &= \frac{KL^2}{\cos^2 \varphi} \left(\frac{\sin \varphi}{\cos \varphi} - \frac{\sin \varphi}{\cos \varphi_0} - \frac{F}{KL} \right) = 0 \Rightarrow \text{study } \frac{F}{KL} = \sin \varphi \left(\frac{1}{\cos \varphi} - \frac{1}{\cos \varphi_0} \right) = \frac{F_{eq}}{KL}(\varphi) \\ &\quad \neq 0 \quad \forall \varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{aligned}$$

Steady $\frac{F}{\kappa L} = \sin\varphi \left(\frac{1}{\cos\varphi} - \frac{1}{\cos\varphi_0} \right) = \frac{F_{eq}}{\kappa L}(\varphi)$

a) Equilibrium for $F=0$: $\varphi=0$, $\cos\varphi = \cos\varphi_0 \Rightarrow \varphi = \pm\varphi_0$ (3 equil. configs)

b) Equilibrium for $F \neq 0$

Expand $\frac{F_{eq}}{\kappa L}(\varphi)$ around $\varphi=0$ and use the fact that $\varphi_0 \ll 1$. In this way $\frac{F_{eq}}{\kappa L}(\varphi)$ will be well approximated in the interval $\varphi \in (-\varphi_0, \varphi_0)$

$$\sin\varphi = \varphi - \frac{\varphi^3}{6!} + \dots$$

$$\frac{1}{\cos\varphi} = 1 + \frac{d}{d\varphi} \frac{1}{\cos\varphi} \Big|_{\varphi=0} + \frac{d^2}{d\varphi^2} \frac{1}{\cos\varphi} \Big|_{\varphi=0} \frac{\varphi^2}{2} + \dots$$

$$\frac{d}{d\varphi} \frac{1}{\cos\varphi} = - \frac{(-\sin\varphi)}{\cos^2\varphi} = \frac{\sin\varphi}{\cos^2\varphi} = 0 \text{ at } \varphi=0$$

$$\frac{d^2}{d\varphi^2} \frac{1}{\cos\varphi} = \frac{d}{d\varphi} \frac{\sin\varphi}{\cos^2\varphi} = \frac{\cos^3\varphi - \sin\varphi 2\cos\varphi(-\sin\varphi)}{\cos^4\varphi} = \frac{\cos^3\varphi + 2\sin^2\varphi\cos\varphi}{\cos^4\varphi} = 1 \text{ at } \varphi=0$$

Therefore $\frac{1}{\cos\varphi} = 1 + \frac{\varphi^2}{2} + \dots$, $\frac{1}{\cos\varphi_0} = 1 + \frac{\varphi_0^2}{2} + \dots$

$$\frac{F_{eq}}{\kappa L}(\varphi) = \left(\varphi - \frac{\varphi^3}{6} + \dots \right) \left(1 + \frac{\varphi^2}{2} + \dots - 1 - \frac{\varphi_0^2}{2} - \dots \right) = \frac{1}{2}(\varphi^2 - \varphi_0^2)\varphi + O(\varphi^4)$$

$\approx$ order three in φ_0, φ

$$\frac{F_{eq}}{\kappa L}(\varphi) \approx \frac{1}{2}(\varphi^2 - \varphi_0^2)\varphi = \frac{1}{2}(\varphi - \varphi_0)(\varphi + \varphi_0)\varphi \quad \text{three zeros: } \varphi=0, \varphi=\pm\varphi_0$$

It's a cubic polynomial. To have the graph

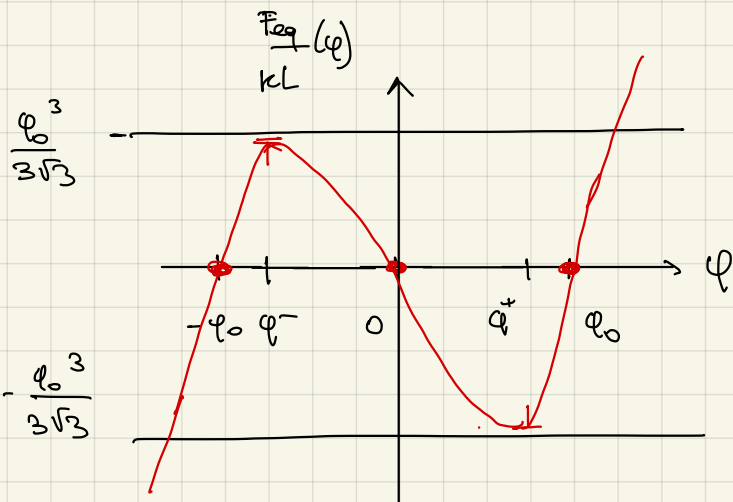
$$\left(\frac{F_{eq}}{\kappa L} \right)' = \frac{d}{d\varphi} \left(\frac{1}{2}\varphi^3 - \frac{1}{2}\varphi_0^2\varphi \right) = \frac{1}{2}(3\varphi^2 - \varphi_0^2) = 0 \Rightarrow \varphi = \pm \frac{\varphi_0}{\sqrt{3}}$$

$$\frac{F_{eq}}{\kappa L} \left(-\frac{\varphi_0}{\sqrt{3}} \right) = \frac{1}{2} \varphi (\varphi^2 - \varphi_0^2) \Big|_{\varphi = -\frac{\varphi_0}{\sqrt{3}}} = \frac{1}{2} \left(-\frac{\varphi_0}{\sqrt{3}} \right) \left(\frac{\varphi_0^2}{3} - \varphi_0^2 \right) = + \frac{\varphi_0^3}{3\sqrt{3}}$$

$\underbrace{\frac{\varphi_0^2}{3} - \varphi_0^2}_{-\frac{2}{3}\varphi_0^2}$

Now can plot the diagram of a cubic polynomial approximation of $\frac{F_{eq}}{\kappa L}(\varphi)$

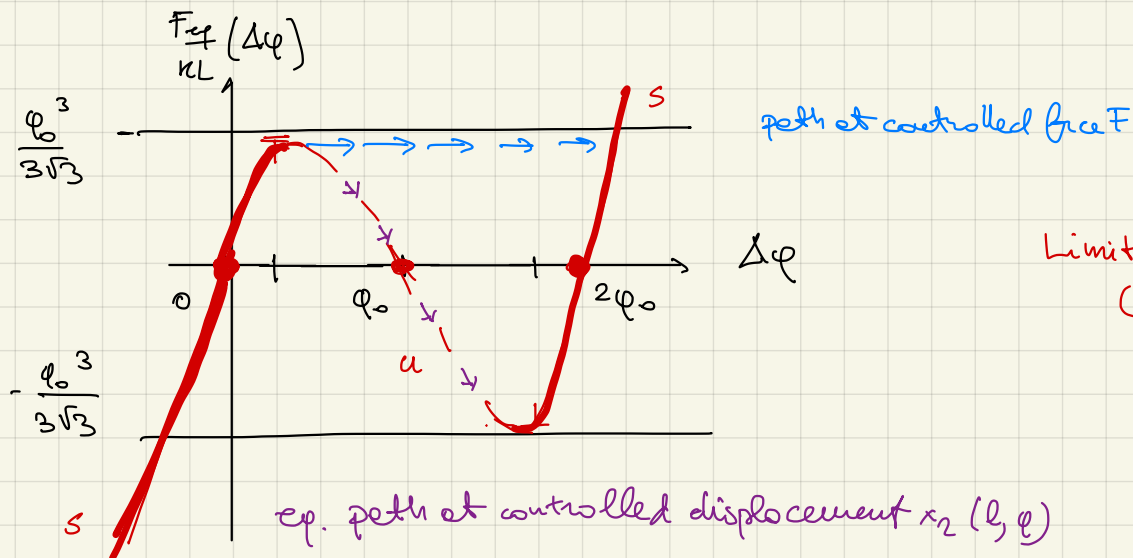
$$\frac{F_{eq}}{k_L}(\varphi) \simeq \frac{1}{2}(\varphi^2 - \varphi_0^2)\varphi = \frac{1}{2}(\varphi - \varphi_0)(\varphi + \varphi_0)\varphi$$



$$\varphi^\pm = \pm \frac{\varphi_0}{\sqrt{3}}$$

It can be shown that the equilibrium configurations $(\varphi, \frac{F_{eq}}{k_L}(\varphi))$ in the graph above are stable for $\varphi \in (-\frac{\pi}{2}, \varphi^-) \cup (\varphi^+, \frac{\pi}{2})$, and unstable for $\varphi \in (\varphi^-, \varphi^+)$

Changing variable from φ to $\Delta\varphi = \varphi - (-\varphi_0) = \varphi + \varphi_0$ we obtain



Limit-point instability (snap-through)

eq. path at controlled displacement $x_2(l, \varphi)$